Clustering-Based Big Data Analysis of Web Crawler Opt-Out Strategies: A Scientific Approach to Robots.txt Governance

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***Abstract*— Large language models (LLMs) rely on large-scale web data collected by automated crawlers, yet their access is increasingly limited by the rules defined in *robots.txt* files. When major websites restrict AI-specific crawlers, the composition of available data for training becomes uneven and less transparent. To analyze this phenomenon quantitatively, this study applies a scientific big-data approach to examine website opt-out strategies across industries. Using the 2024 BEDC list of the world's top 100 websites, 98 valid *robots.txt* files were automatically collected, standardized, and analyzed through K-means clustering and principal component analysis (PCA). The results identify three dominant blocking strategies—full opt-out, partial AI-targeted opt-out, and rate-limited access—and show that 61.2% of the sampled websites block at least one AI crawler, indicating a measurable contraction of open web data. PCA further reveals two underlying governance dimensions: blocking intensity and pattern differentiation, which explain how websites translate policy choices into distinct forms of data control. The proposed analytical framework provides a data-driven foundation for studying large-scale web accessibility and demonstrates how big-data analytics can uncover emerging patterns in digital governance and information control.**

Keywords—big data analytics, clustering, robots.txt, AI crawler, data accessibility, data governance

I. INTRODUCTION

Large language models (LLMs) are increasingly used in search, content generation, and decision-making tasks, yet their factual reliability remains uncertain. Because these models depend on large-scale web data collected by automated crawlers [10], any restriction defined in *robots.txt* can shape what information becomes part of their training corpus. As generative AI expands, *robots.txt* has evolved from a conventional search-engine protocol into a key mechanism governing AI data access, with many websites now blocking AI-specific crawlers.

In recent years, the question of data accessibility for AI systems has gained growing importance. The rapid emergence of AI crawlers such as GPTBot and ClaudeBot has made robots.txt a visible policy instrument that determines which parts of the web remain open to machine learning models. This shift reflects a broader transformation of web infrastructure, moving from open indexing for search to selective data sharing for AI. A data-driven, cross-industry perspective is therefore essential to understand how web openness and control are evolving on a global scale.

An analysis by Stox and Guan (2025), based on the Ahrefs database of roughly 140 million websites, shows that GPTBot is the most frequently blocked crawler, followed by ClaudeBot and Google-Extended [9]. This trend highlights rising concern over data ownership and the reluctance of web publishers to share content with AI systems. Such selective blocking can narrow the diversity of training data [5], producing coverage gaps that increase the risk of factual inconsistency and reduce public trust [6].

Most prior studies have focused on traditional crawler behaviors [1] or the legal and ethical dimensions of web access such as copyright and data responsibility [2], often limited to specific sectors or regions [3]. However, only a few recent works have attempted to examine AI-targeted access policies at scale [17][18]. To address this research gap, the present study develops a clustering-based big data framework that converts *robots.txt* rules into structured analytical features and applies k-means clustering to identify patterns of AI crawler restriction across major websites.

The novelty of this research lies in translating a largely qualitative governance topic into a quantitative, scientific framework that uses big data analytics and visualization to uncover the structure of web accessibility. The findings provide new insight into how data coverage and access restrictions shape the information environment of AI training. This work contributes to the ICBDA theme of Scientific Applications of Big Data by demonstrating how large-scale web metadata can be systematically analyzed through clustering and visualization techniques to reveal underlying patterns of accessibility and control in the digital ecosystem.

II. LITERATURE REVIEW

A. The Role of Robots.txt in Internet Governance

Early discussions of internet governance largely revolved around the management of search engines [14]. Sun, Zhuang, and Giles [1], who analyzed millions of websites, pointed out that the robots.txt protocol was first introduced as a form of self-regulation, designed to help websites coordinate with crawlers and avoid overloading their servers.

As generative AI applications have emerged, the purpose of robots.txt has shifted. What began as a technical standard

now carries the weight of content ownership and data governance [2][3]. Liu et al. (2025) [3] note that many artists use robots.txt to keep AI crawlers away from their work, hoping to prevent their creations from being absorbed into training data. This refusal represents a growing sense of agency among creators in deciding how their digital content circulates [16]. Likewise, Choe et al. (2024) [4] observe that AI models depend heavily on user-generated content, yet the people who provide such data rarely receive credit or compensation. The result is a widening gap of benefit and control between those who produce data and those who build models.

In this light, robots.txt has evolved beyond a scheduling tool for search engines into a symbol of governance in the age of artificial intelligence. It expresses not only a technical preference but also a position on the flow of information itself. Still, most existing work remains focused on legal or ethical debates [14][15]. Quantitative, cross-sector analysis of how websites implement these restrictions remains limited, a gap that this study seeks to address.

B. Dataset Accessibility and LLM Hallucinations

The robots.txt protocol was created to regulate the behavior of web crawlers [1]. It functions like a "no entry" sign at a website's gate, determining what information can or cannot be accessed [3]. As Yang et al. (2024) [11] argue, the move toward closed-sourcing in large language models (LLMs) has limited researchers' ability to examine their underlying data and architecture. This observation underscores how crucial data accessibility is for understanding and improving these systems.

Accessibility also shapes the completeness and representativeness of training corpora. Studies show that when training data are skewed or incomplete, models are more prone to hallucination. Huang et al. (2025) [13] categorize hallucinations into two types: those caused by insufficient contextual understanding and those stemming from data scarcity. Bang et al. (2023) [12] further highlight that hallucination often correlates with sparse or unevenly distributed datasets, reinforcing the importance of comprehensive web data for LLM development.

Taken together, robots.txt serves as a practical "gatekeeper" of data visibility in the LLM ecosystem. When a large number of websites deny access to AI crawlers, the resulting structural gaps in training data can lower model accuracy and heighten hallucination risks, particularly in high-stakes applications [10].

C. Need for a Cross-Industry Typology and Data Source

To link the discussion of governance with the risks of hallucination, this study introduces a practical framework for comparing how different industries manage AI crawlers. We extracted eight structured features from each robots.txt file, covering aspects such as blocking scope and intensity, AI-specific restrictions, delay control, the use of wildcards, overlap with search engine rules, transparency, and the disclosure of sitemaps. Together, these variables provide a picture of how websites across sectors define the boundaries of data access.

By examining these features, we build a quantitative typology of AI user-agent opt-out behaviors that helps us recognize distinct blocking patterns and strategic preferences. The analysis draws on the 2024 BEDC list of the world's top 100 high-traffic websites and applies K-means clustering along with principal component analysis to group the observed behaviors into three broad strategies.

This typology shows that AI crawler restrictions are not uniform but shaped by industrial context and policy attitudes toward data sharing. It also offers an empirical basis for exploring how selective blocking might influence data coverage, model bias, and the broader ecology of AI training. In this way, the study uses quantitative analysis to bridge the gap between governance theory and real-world practice, suggesting how data policy might evolve in the age of generative AI.

III. RESEARCH METHOD

A. Data Collection

This study draws on the 2024 BEDC framework, which lists the world's top 100 high-traffic websites [7][8]. The sample spans news, business, government, and social platforms, offering broad industry coverage. *robots.txt* files were retrieved automatically using Python. A few Microsoft domains, such as *office.com* and *outlook.com*, could not be accessed due to permission limits and were excluded. In total, 98 valid samples were collected.

Because the BEDC list reflects global web traffic, the dataset captures the main policy orientations of leading content providers and serves as a solid basis for cross-industry analysis.

The study adopts a four-stage analytical pipeline encompassing data collection, feature extraction, clustering, and visualization. Python scripts automatically retrieved *robots.txt* files from the 2024 BEDC dataset. The files were parsed into structured variables, standardized, and clustered using the k-means algorithm. Finally, principal component analysis (PCA) was applied to visualize the relationships among clusters. Figure 1 outlines the workflow.

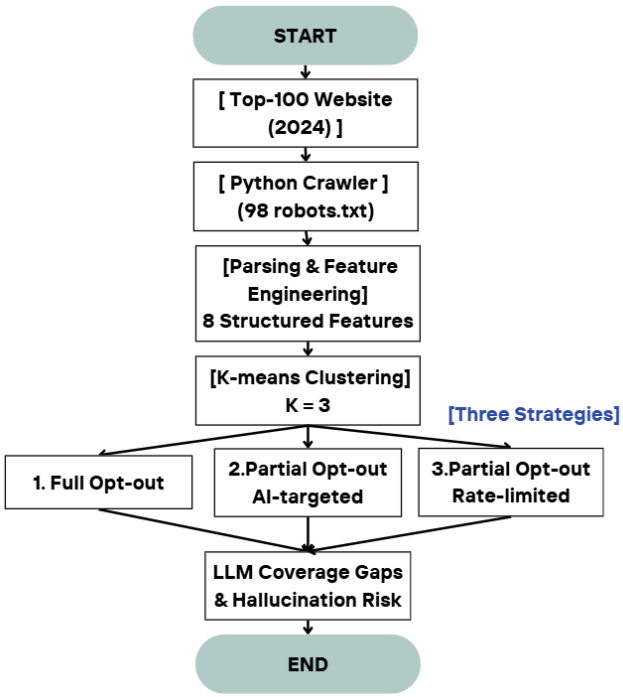

Fig. 1. Workflow of the clustering-based big data analysis, from data collection to PCA visualization.

B. Feature Engineering

Each robots.txt file was parsed by the User-Agent block to extract key rules such as Disallow, Allow, Crawl-delay, and Sitemap. All commands were converted to lowercase, comments were removed, and spaces normalized. Regular expressions handled wildcards (*) and end markers ($) for consistent interpretation. When multiple blocks referred to the same user agent, their rules were merged. Conflicts were resolved by giving priority to more specific matches.

Table 1: Engineered robots.txt features for clustering

Feature	Type	Description
AI_block_count	Integer	Count of explicitly disallowed AI user-agents (e.g., GPTBot, ClaudeBot).
Block_search_engines	Binary {0,1}	Whether major search crawlers (e.g., Googlebot, Bingbot) are also restricted.
Global_disallow_root	Binary {0,1}	Presence of a global Disallow: / directive (site-wide block).
Disallow_scope_ratio	Continuous	Share of disallowed directories relative to a fixed representative set (denominator = 30).
Crawl_delay_norm	Continuous	Normalized Crawl-delay reflecting rate-limiting policies.
AI_only_block	Binary {0,1}	Restrictions apply exclusively to AI crawlers while search engines remain permitted.
Wildcard_usage	Binary {0,1}	Presence of wildcard characters (*, $) in rules, indicating generalized patterns.
Sitemap_present	Binary {0,1}	One or more Sitemap files declared in robots.txt.

Table 2: Representative Directory List (n = 30)

Category	Example Directories
Navigation & Info	*/about, /contact, /search, /blog, /news*
Policy & Trust	*/privacy, /terms, /help, /support*
Account & Auth	*/login, /logout, /signup, /account, /settings*
Commerce & Content	*/store, /products, /pricing, /cart, /category*
Developer & Platform	*/api, /developers, /docs*
Media & Feeds	*/images, /videos, /static, /media, /sitemap.xml*
Community & Careers	*/community, /jobs, /press*

All binary variables were encoded as {0,1}. When *Crawl-delay* or *Sitemap* was not defined in a robots.txt file, a value of 0 was assigned to keep feature dimensions consistent. Before clustering, all variables were standardized with *StandardScaler* so that each feature had a mean of 0 and a standard deviation of 1, avoiding scale bias.

For reproducibility, the AI user-agent dictionary and representative directory list used to compute *disallow_scope_ratio* were fixed throughout the analysis. The denominator contained 30 directories (Table 2), chosen to balance coverage and stability by reflecting common website structures such as navigation, policy, account, and commerce paths.

C. Clustering Procedure

To determine the appropriate number of clusters, we adopted the elbow method, which evaluates the trade-off between model complexity and explanatory power. For each candidate value of k (ranging from 2 to 10), the k-means algorithm was repeated using the same initialization settings, and the within-cluster sum of squared errors (SSE) was recorded. The SSE measures how tightly the data points in each cluster are grouped around their respective centroids;

smaller values indicate more compact and well-separated clusters.

As shown in Figure 2, the SSE decreased sharply as k increased from 2 to 3, but the rate of improvement gradually flattened beyond that point. This turning point, commonly referred to as the "elbow," marks the stage where adding more clusters yields diminishing returns in reducing within-cluster variance. Based on this observation, k = 3 was selected as the optimal configuration, balancing interpretability with statistical compactness. The resulting clusters were then used for further comparison and visualization.

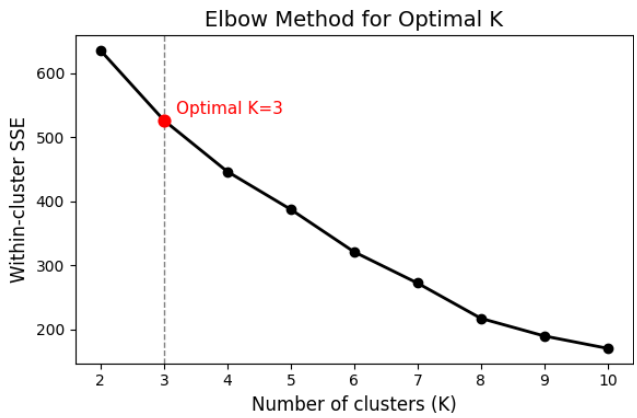

Fig. 2. Elbow curve showing the relationship between the number of clusters (k) and the within-cluster sum of squared errors (SSE). The elbow at $k = 3$ represents the point where additional clusters bring minimal improvement in fit.

IV. Results and Discussion

A. Overview

Using eight standardized features, the K-means analysis (K = 3) grouped the sampled websites into three stable clusters. The PCA visualization (Figure 3) shows clear separation and distinct centroids among the clusters, corresponding to three blocking strategies: *Full Opt-out*, *Partial Opt-out (AI-targeted)*, and *Partial Opt-out (Rate-limited)*.

The clusters included 43 websites in the *Full Opt-out* group, 38 in the *Rate-limited* group, and 17 in the *AI-targeted* group. Overall, about 61.2 percent of the sampled websites applied some form of restriction to AI crawlers. This trend reflects a growing tendency for data owners to exercise control over how their content is used, signaling that AI access management is becoming an integral part of data governance.

As discussed in Section II-B on dataset accessibility and LLM hallucinations, widespread blocking of AI crawlers could lead to uneven data coverage and gaps in training corpora. Such imbalances may reduce model accuracy and raise the risk of hallucinations, underscoring the need for more transparent and balanced approaches to web data governance.

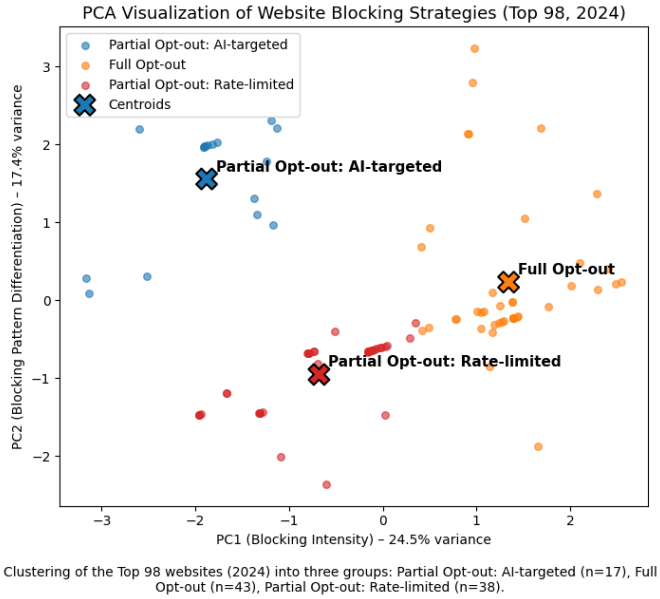

Fig. 3. PCA visualization of website blocking strategies among the Top 98 websites (2024). PC1 (Blocking Intensity) explains 24.5% of variance, and PC2 (Blocking Pattern Differentiation) explains 17.4%, together accounting for 41.9% of total variance.

B. Interpretation of Principal Components (PC1 and PC2)

To clarify the dimensions underlying the clustering analysis, the two principal components obtained from the PCA visualization (Figure 3) are examined in this section. Each component represents a latent construct derived from the standardized feature matrix through eigen-decomposition of the covariance matrix, summarizing multiple correlated variables into orthogonal dimensions. The cumulative variance was obtained by summing the explained-variance ratios of the first two components, which were calculated from the eigenvalues of the covariance matrix of the standardized features. Specifically, PC1 accounted for 24.5% and PC2 for 17.4% of the total variance, giving a combined value of 41.9%. These ratios were derived directly from the PCA output in scikit-learn and confirm that the first two components capture the dominant structural variation in the dataset. Together, PC1 and PC2 explain 41.9% of the total variance in website blocking behaviors and provide a conceptual basis for interpreting their governance orientations.

PC1 (Blocking Intensity): The first principal component captures the overall restrictiveness and comprehensiveness of blocking policies. Websites with high PC1 values tend to adopt stringent rules characterized by site-wide disallow directives, extensive wildcard usage, and large disallow-scope ratios. They also frequently restrict major search crawlers such as Googlebot or Bingbot. These characteristics indicate a defensive or exclusion-oriented approach to data governance, emphasizing content control and limited external visibility.

Conversely, websites with low PC1 values exhibit more permissive configurations. Their robots.txt files contain fewer disallow entries, allowing broader crawler access and reflecting a visibility- and accessibility-oriented stance. Overall, PC1 distinguishes between comprehensive restriction and openness, and can thus be interpreted as an indicator of blocking intensity.

Dominant variables:

1.G*lobal_disallow_root*	*2.Disallow_scope_ratio*
3.Block_search_engines	*4.Wildcard_usage*

PC2 (Blocking Pattern Differentiation): The second principal component represents the degree of differentiation in blocking patterns, particularly the extent to which a website distinguishes between conventional search crawlers and AI-specific agents. Websites with high PC2 scores exhibit selective restrictions that explicitly identify and block AI-related user agents (e.g., GPTBot, ClaudeBot) while maintaining access for general-purpose search engines. This selective configuration, characterized by high *AI_block_count* and *AI_only_block* values, reflects a targeted and differentiated blocking strategy motivated by concerns over content reuse or proprietary data control.

In contrast, low PC2 values correspond to uniform or technically motivated restrictions, such as rate-limiting directives (*Crawl_delay_norm*) applied across all crawlers without distinction. These configurations prioritize operational efficiency and server protection rather than content governance per se. Hence, PC2 captures blocking pattern differentiation, distinguishing between selective, policy-driven restriction and generic, performance-oriented control.

Dominant variables:

1.AI_block_count	*2.AI_only_block*
3.Crawl_delay_norm	

In sum, the horizontal axis (PC1) reflects the extent of website restriction, whereas the vertical axis (PC2) reflects the nature or selectivity of that restriction. The two components jointly form a conceptual space in which blocking strategies can be positioned according to their degree of openness, differentiation, and operational emphasis. Beyond visualization, PCA functions as a scientific big-data analysis tool by converting multidimensional *robots.txt* variables into interpretable governance dimensions. It enables the discovery of latent behavioral structures within large-scale datasets and provides a quantitative basis for comparing how different sectors balance openness, control, and technical management in web accessibility.

C. Key Finding

The three clusters reveal distinct levels of restriction and policy orientation, aligning with the ICBDA theme of *Scientific Applications of Big Data*. This differentiation illustrates how quantitative big-data techniques can uncover structural patterns in web accessibility and governance behavior. SMD (standardized mean difference) was used to measure feature-level variation among clusters, with higher SMD values (≥ 0.8) indicating substantial differences. This provides a consistent, statistical basis for interpreting how websites diverge in their governance strategies. Detailed statistics for each cluster are available in the project's public GitHub repository [8].

(1) Full Opt-out. This cluster reflects the most defensive and comprehensive strategy. These websites make extensive use of the *Disallow* directive and wildcards to maximize coverage (*Wildcard_usage mean* = 1.000, SMD = 0.787). They also restrict nearly all major search engines (*Blocks_search_engines* = 0.977, *SMD* = 1.974). The high *Disallow_scope_ratio* (mean = 9.493, SMD = 0.953) confirms a governance stance centered on control and exclusion. This group seeks to eliminate crawler visibility altogether, effectively removing its content from both AI and search ecosystems.

(2) Partial Opt-out: AI-targeted. This cluster demonstrates a selective and strategically adaptive approach. Websites in this group generally allow search engines but explicitly restrict AI-specific crawlers such as GPTBot and ClaudeBot. It records the highest *AI_block_count* (mean = 4.941, SMD = 1.074) and complete *AI_only_block* (mean = 1.000, SMD = 2.641). These characteristics indicate a targeted governance model, where access control is redefined not by the nature of the crawler's function (search vs. scrape) but by its intended use in generative AI. This pattern marks a transition from search exposure management toward protecting proprietary or creative data assets from AI reuse.

(3) Partial Opt-out: Rate-limited. This cluster represents a pragmatic and performance-oriented stance. Websites in this group set moderate *crawl-delay* directives (mean = 0.553, SMD = 0.163) and maintain a smaller *Disallow_scope_ratio* (mean = 1.595). Rather than blocking access outright, they implement technical throttling mechanisms to ensure operational stability and bandwidth efficiency. This cluster therefore exemplifies a functional governance logic, where accessibility is maintained but regulated through speed and frequency constraints. It reflects a middle ground between openness and system self-protection.

Across clusters, the SMD results reveal a gradient of restriction, ranging from comprehensive exclusion (Full Opt-out) to selective AI filtering (Partial AI-targeted) and finally to operational control (Rate-limited). This gradient corresponds to the PCA-derived dimensions: PC1 (Blocking Intensity) and PC2 (Pattern Differentiation), jointly explaining how platforms translate policy choices into measurable behavioral patterns.

D. ***Discussion***

Building on the clustering results, this section explores how these strategies reflect broader shifts in AI data governance and what they mean for model transparency and accountability.

The findings show that the focus of website data governance is changing. Robots.txt, once a simple rule for search engine indexing, has evolved into a mechanism for deciding how much access AI crawlers should have. The space of open data available for model training is gradually narrowing, and this quiet shift could lead to uneven coverage and domain bias that undermine the fairness and reliability of large language models.

The clustering analysis also reveals clear differences in how websites interpret and apply AI access policies. The variation in SMD values indicates that openness is not distributed evenly but shaped by each platform's policy priorities and institutional goals. Taken together, these findings suggest that data governance is no longer a purely technical matter. It has become a negotiation of power and responsibility over who defines, owns, and controls the knowledge that AI systems learn from..

V. CONCLUSION

About 61.2 percent of the websites in our sample block at least one AI crawler. This finding reflects a clear shift in data governance, moving from managing search engine visibility to determining the extent of access granted to AI systems. The amount of open web data available for training large language models is already shrinking, and this contraction is likely to accelerate as more websites adopt AI-targeted restrictions.

Among all factors, the breadth and strictness of *Disallow* rules exert the greatest influence on data accessibility. The variation in SMD values across clusters confirms distinct policy orientations among websites, corresponding to three main strategies identified by the clustering analysis: full opt-out, partial AI-targeted opt-out, and rate-limited access. These clusters align with the PCA-derived dimensions of blocking intensity (PC1) and pattern differentiation (PC2), revealing how policy choices translate into measurable structural differences. By building a reproducible, feature-based model from *robots.txt* files, this study provides a concrete, data-driven method for observing how these governance patterns emerge in practice. It addresses a gap left by earlier research that focused primarily on law or ethics rather than on empirical data.

Robots.txt is no longer just a technical note for crawlers; it has become part of the policy language of the generative AI era, serving as a public signal of how data holders negotiate openness and control. As AI-blocking rules spread, the equilibrium of global open data will continue to shift, shaping what models can learn and influencing the reliability and representativeness of their knowledge.

Overall, this study presents a new analytical framework for examining large-scale web accessibility through big-data methods. By integrating automated data collection, clustering, and principal component analysis (PCA), it demonstrates how scientific analytics can uncover latent structures in online information control. This approach not only deepens our understanding of web-based data governance but also provides a scalable foundation for future research on transparency, accountability, and the sustainability of digital ecosystems.

REFERENCES

[1] Sun, Y., Zhuang, Z., & Giles, C. L. (2007, May). A large-scale study of robots. txt. In Proceedings of the 16th international conference on World Wide Web (pp. 1123-1124).

[2] Chang, C. Y., & He, X. (2025). The Liabilities of Robots. txt. arXiv preprint arXiv:2503.06035.

[3] Liu, E., Luo, E., Shan, S., Voelker, G. M., Zhao, B. Y., & Savage, S. (2024). Somesite i used to crawl: Awareness, agency and efficacy in protecting content creators from ai crawlers. arXiv preprint arXiv:2411.15091.

[4] Choe, S. K., Ahn, H., Bae, J., Zhao, K., Kang, M., Chung, Y., ... & Xing, E. (2024). What is your data worth to gpt? llm-scale data valuation with influence functions. arXiv preprint arXiv:2405.13954.

[5] Parrots, S. (2021). Can Language Models Be Too Big. Proceedings of the 2021 ACM.

[6] Chen, C., & Shu, K. (2023). Can llm-generated misinformation be detected?. arXiv preprint arXiv:2309.13788.

[7] Cheng, C. Y. (2025). Determinants of firms' decisions to block AI crawlers: Strategic insights from the BEDC framework. Technological Forecasting and Social Change, 219, 124275.

[8] Cheng, B.-R., & Cheng, C.-Y. (2024). Robots.txt opt-out dataset (BEDC Top-100, 2024) [Data set]. GitHub. https://github.com/Raycheng0629/robots_txt-optout-dataset

[9] Stox, P., & Guan, X. (2025, May 21). The AI bots that ~140 million websites block the most. Ahrefs Blog. https://ahrefs.com/blog/ai-bot-block-rates/

[10] Fan, D., Sabolčec, V., Ansaripour, M., Tarun, A. K., Jaggi, M., Bosselut, A., & Schlag, I. (2025). Can Performant LLMs Be Ethical? Quantifying the Impact of Web Crawling Opt-Outs. arXiv preprint arXiv:2504.06219.

[11] Yang, J., Jin, H., Tang, R., Han, X., Feng, Q., Jiang, H., ... & Hu, X. (2024). Harnessing the power of llms in practice: A survey on chatgpt and beyond. ACM Transactions on Knowledge Discovery from Data, 18(6), 1-32.

[12] Bang, Y., Cahyawijaya, S., Lee, N., Dai, W., Su, D., Wilie, B., ... & Fung, P. (2023). A multitask, multilingual, multimodal evaluation of chatgpt on reasoning, hallucination, and interactivity. arXiv preprint arXiv:2302.04023.

[13] Huang, L., Yu, W., Ma, W., Zhong, W., Feng, Z., Wang, H., ... & Liu, T. (2025). A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. ACM Transactions on Information Systems, 43(2), 1-55.

[14] Novelli, C., Taddeo, M., & Floridi, L. (2024). Accountability in artificial intelligence: What it is and how it works. Ai & Society, 39(4), 1871-1882.

[15] Cath, C. (2018). Governing artificial intelligence: ethical, legal and technical opportunities and challenges. Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences, 376(2133), 20180080.

[16] Floridi, L., & Cowls, J. (2022). A unified framework of five principles for AI in society. Machine learning and the city: Applications in architecture and urban design, 535-545.

[17] Kim, T., Bock, K., Luo, C., Liswood, A., & Wenger, E. (2025). Scrapers selectively respect robots. txt directives: evidence from a large-scale empirical study. arXiv preprint arXiv:2505.21733.

[18] Longpre, S., Mahari, R., Lee, A., Lund, C., Oderinwale, H., Brannon, W., ... & Pentland, A. (2024). Consent in crisis: The rapid decline of the ai data commons. Advances in Neural Information Processing Systems, 37, 108042-108087.